

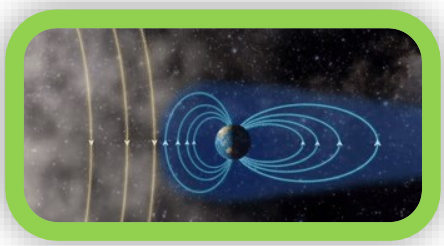
EP2T # 8

MAGNETIC FIELD AND MAGNETIC FORCES

&

SOURCES OF MAGNETIC FIELD

Lecture 8 & 9





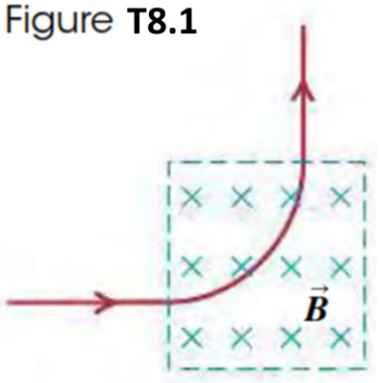
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Question 1:

A beam of protons traveling at 1.20 km/s enters a uniform magnetic field, traveling perpendicular to the field. The beam exits the magnetic field, leaving the field in a direction perpendicular to its original direction (**Fig. T8.1**). The beam travels a distance of 1.18 cm *while in the field*. What is the magnitude of the magnetic field?

Figure **T8.1**



Question 2:

A singly charged ion of ^{7}Li (an isotope of lithium) has a mass of 1.16×10^{-26} kg. It is accelerated through a potential difference of 220 V and then enters a magnetic field with magnitude 0.874 T perpendicular to the path of the ion. What is the radius of the ion's path in the magnetic field?

Question 3:

Singly ionized (one electron removed) atoms are accelerated and then passed through a velocity selector consisting of perpendicular electric and magnetic fields. The electric field is 155 V/m and the magnetic field is 0.0315 T. The ions next enter a uniform magnetic field of magnitude 0.0175 T that is oriented perpendicular to their velocity. (a) How fast are the ions moving when they emerge from the velocity selector? (b) If the radius of the path of the ions in the second magnetic field is 17.5 cm, what is their mass?

Question 4:

A particle with charge 7.80 μC is moving with velocity $\vec{v} = -(3.80 \times 10^3 \text{m/s})\hat{j}$. The magnetic force on the particle is measured to be $\vec{F} = +(7.60 \times 10^{-3}\ \text{N})\hat{i} - (5.20 \times 10^{-3}\ \text{N})\hat{k}$. (a) Calculate all the components of the magnetic field you can from this information. (b) Are there components of the magnetic field that are not determined by the measurement of the force? Explain. (c) Calculate the scalar product $\vec{B} \cdot \vec{F}$. What is the angle between $\vec{B}$ and $\vec{F}$?

Question 5:

A square wire loop 10.0 cm on each side carries a clockwise current of 8.00 A. Find the magnitude and direction of the magnetic field at its center due to the four 1.20-mm wire segments at the midpoint of each side.

Question 6:

The wires in a household lamp cord are typically 3.0 mm apart center to center and carry equal currents in opposite directions. If the cord carries direct current to a 100-W light bulb connected across a 120-V potential difference, what force per meter does each wire of the cord exert on the other? Is the force attractive or repulsive? Is this force large enough so it should be considered in the design of the lamp cord? (Model the lamp cord as a very long straight wire.)

Question 7:

As a new electrical technician, you are designing a large solenoid to produce a uniform 0.150-T magnetic field near the center of the solenoid. You have enough wire for 4000 circular turns. This solenoid must be 55.0 cm long and 2.80 cm in diameter. What current will you need to produce the necessary field?

Question 8:

An alpha particle (charge +2*e*) and an electron move in opposite directions from the same point, each with the speed of 2.50 × 105 m/s (**Fig. T8.2**). Find the magnitude and direction of the total magnetic field these charges produce at point *P*, which is 8.65 nm from each charge.

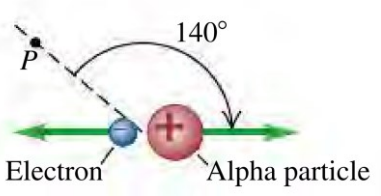


Fig. T8.2

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Extra Questions for practice:

Question 9:

The plane of a 5.0 cm × 8.0 cm rectangular loop of wire is parallel to a 0.19-T magnetic field. The loop carries a current of 6.2 A.

(a) What torque acts on the loop?

(b) What is the magnetic moment of the loop?

(c) What is the maximum torque that can be obtained with the same total length of wire carrying the same current in this magnetic field?

Question 10:

Calculate the magnetic field (magnitude and direction) at a point *P* due to a current *I* = 12.0 A in the wire shown in **Fig. T8.3**. Segment *BC* is an arc of a circle with radius 30.0 cm, and point *P* is at the center of curvature of the arc. Segment *DA* is an arc of a circle with radius 20.0 cm, and point *P* is at its center of curvature. Segments *CD* and *AB* are straight lines of length 10.0 cm each

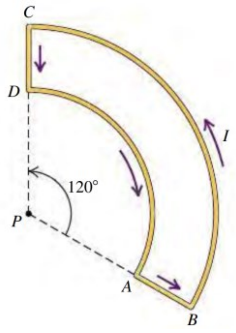


Fig. T8.3

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